

Individual Case Study

Topic: "The Shrink Before the Hike": Detecting Silent Shrinkflation in Indian Packaged Grocery Products Using Web-Scraped Price-and-Weight Trends

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Course Name: Business Analytics

Executive Summary

Shrinkflation is a pricing practice in which the quantity of a packaged product is reduced while the displayed price is maintained or changed by less than the change in quantity. This case study examines the idea through a longitudinal price-and-pack-size analysis of Indian packaged grocery products. The project converts product prices and package quantities into a common unit-price measure, compares baseline and terminal observations, separates price changes from quantity changes, and then applies classification models to identify product characteristics associated with downsizing.

The supplied analytical dataset contains 16,800 observations for 1,400 unique SKUs across 12 quarterly waves from 2022-Q1 to 2024-Q4. In the current prototype dataset, 960 SKUs (68.6%) meet the project's shrinkflation rule of a pack-weight reduction greater than 3%. Among these flagged SKUs, the mean weight change is -21.8%, while mean unit-price change across the full SKU set is 31.9%.

Low-unit-pack products priced at ₹5, ₹10 or ₹20 show a simulated shrinkflation incidence of 95.1%, compared with 45.2% for other products. The four-quadrant analysis classifies 45.4% of SKUs as pure silent shrink, 23.1% as double-whammy cases, 24.6% as overt inflation and 6.8% as stable/fair-value cases.

Three supervised models—Logistic Regression, Random Forest and Gradient Boosting—were evaluated. All three produce perfect holdout metrics in the current simulated dataset. This result is treated cautiously: the dataset is generated rather than independently observed, and the target definition is closely related to the price/weight features used in the analysis. The models are therefore useful for demonstrating the analytical pipeline, but their reported performance should not be interpreted as evidence of real-world predictive accuracy.

Measure	Current project value
Unique SKUs	1,400
Observation records	16,800
Observation waves	12
Period	2022-Q1 to 2024-Q4
Flagged shrinkflation SKUs	960 (68.6%)
Mean weight change among flagged SKUs	-21.8%
Mean unit-price change	31.9%
₹5/₹10/₹20 shrinkflation incidence	95.1%

1. Problem Statement and Objectives

1.1 Problem Statement

Consumers usually notice an increase in the printed price of a grocery product more easily than a small reduction in its package quantity. For products sold at familiar low unit prices, such as ₹5, ₹10 and ₹20, a manufacturer may have an incentive to adjust the quantity instead of making the headline price change. The result is that the price per gram or millilitre can increase even when the displayed price appears stable.

The business analytics problem is therefore to move beyond the sticker price and measure the effective price paid for a standard quantity. The study asks whether repeated reductions in pack size are associated with higher unit prices and whether simple product characteristics can help identify products that are more likely to be downsized.

1.2 Objectives

1. Measure changes in standardised price per 100g or 100ml between the first and last observation waves.
2. Separate effective unit-price changes into the contribution of nominal price changes and package-size changes.
3. Classify products into overt inflation, silent shrinkflation, double-whammy and stable/fair-value groups.
4. Evaluate Logistic Regression, Random Forest and Gradient Boosting for shrinkflation classification.

- Construct a project-specific Brand Transparency Index (BTI) to summarise product-level patterns at company level.
- Translate the findings into practical recommendations for pricing, product communication and consumer comparison.

2. Data Collection and Dataset

2.1 Dataset status and provenance

The dataset is nevertheless structured in the same form required by a longitudinal web-scraping study. It contains product identifiers, product names, brands, categories, observation dates, platform, city zone, MRP, selling price, package quantity, unit price and product-attribute fields. This makes it suitable for testing the complete analytics workflow before real scraped observations are inserted.

2.2 Dataset scope

Attribute	Value
Unique SKUs	1,400
Observations	16,800
Waves	12
Time period	2022-Q1 to 2024-Q4
Platforms represented	JioMart, Blinkit, Zepto, BigBasket
City zones represented	Delhi NCR, Mumbai, Kolkata, Chennai, Bengaluru, Hyderabad
Product categories	6
Observations per SKU	12

2.3 Data fields

Field group	Examples
Product identity	observation_id, sku_id, product_name, brand
Classification	parent_company, category, sub_category
Time	wave_id, period_label, observation_date
Market context	source_platform, city_zone
Pricing	mrp_inr, selling_price_inr, unit_price_per_100
Quantity	pack_weight_volume, unit_of_measure
Price-point indicators	is_magic_price_point, magic_price_tier
Packaging/claims	packaging_type, marketing_redesign_claim
Generator label	intended_behavior_archetype

2.4 Intended web-scraping workflow

For the final course-compliant version, the same schema should be populated from publicly accessible product pages on platforms such as Blinkit, Zepto, BigBasket and JioMart. A real collection run should record the page URL, product identity, package size, MRP/selling price and collection timestamp, preserve the raw snapshot, and never invent historical dates. Repeated runs create the longitudinal panel.

- Open a publicly accessible category or product page.
- Extract product name, brand, category, package size and price.
- Normalise units such as kg→g and L→ml.
- Add the actual collection date and time.

- Save each run as an immutable raw snapshot.
- Match the same SKU across later snapshots and calculate changes.

3. Data Preparation and Exploratory Analysis

3.1 Preparation

The analytical file is built at SKU level by sorting observations chronologically and comparing the first and last waves. Prices and package quantities are standardised so that unit-price comparisons are meaningful across products with different pack sizes.

- Standardised package quantities into a consistent numerical measure.
- Calculated price per 100g or 100ml.
- Matched each SKU's baseline and terminal observations.
- Calculated percentage changes in nominal price, package quantity and unit price.
- Applied a project-defined 3% weight-reduction threshold for shrinkflation classification.
- Encoded category and packaging variables for supervised learning.

Threshold clarification

The 3% threshold is a project classification rule used to identify meaningful pack-size reduction. It is not presented as a legal definition or legal tolerance for shrinkflation.

3.2 Category-level results

Category	SKUs	Shrunk	Shrink rate	Mean weight change	Mean unit-price change
Snacks & Instant Foods	309	251	81.2%	-17.6%	+33.2%
Biscuits & Bakery	386	311	80.6%	-17.6%	+33.9%
Personal Care & Soaps	263	197	74.9%	-16.6%	+33.2%
Dairy & Breakfast	153	93	60.8%	-12.6%	+31.0%
Home Care & Cleaning	154	83	53.9%	-12.3%	+29.5%
Cooking Staples & Condiments	135	25	18.5%	-4.0%	+24.2%

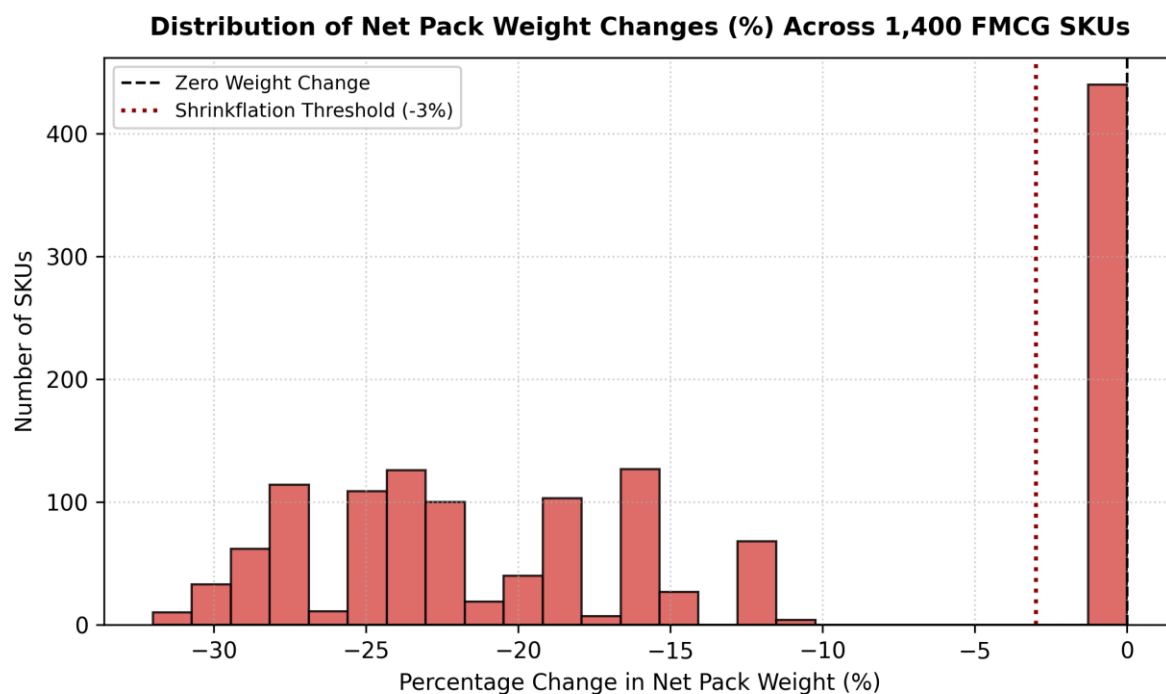


Figure 1. Distribution of baseline-to-terminal package-weight changes in the analytical prototype dataset.

The simulated data show the highest shrinkflation incidence in Snacks & Instant Foods (81.2%) and Biscuits & Bakery (80.6%). Cooking Staples & Condiments shows the lowest rate (18.5%). These differences are descriptive patterns within the generated dataset and should not be interpreted as market estimates.

4. Analytics Method and Implementation

4.1 Standardised unit price

For solid products:

$$\text{Unit Price} = (\text{Selling Price} / \text{Package Weight}) \times 100$$

For liquid products, the same idea is applied using millilitres as the quantity unit.

4.2 Log-differenced decomposition

The project uses the identity $U = P/W$, where U is unit price, P is nominal price and W is package quantity. Taking logarithms gives:

$$\Delta \ln(U) = \Delta \ln(P) - \Delta \ln(W)$$

This decomposition makes the two channels visible. A positive $\Delta \ln(P)$ represents a contribution from the sticker-price change, while a negative $\Delta \ln(W)$ increases unit price through quantity reduction.

4.3 Stealth Index

For products with positive unit-price growth, the project calculates a bounded Stealth Index (SI) as the share of the unit-price increase associated with the package-weight reduction:

$$SI = \max(0, \min(1, -\Delta\%W / \Delta\%U))$$

4.4 Four-quadrant classification

Group	Rule	Current prototype share
Q1 – Overt Inflation	Weight change $\geq -3\%$; price change $> 3\%$	24.6%

Q2 – Silent Shrink	Weight change < -3%; price change ≤ 3%	45.4%
Q3 – Double Whammy	Weight change < -3%; price change > 3%	23.1%
Q4 – Stable/Fair	Neither threshold is crossed	6.8%

Distribution of 1,400 Grocery SKUs Across Pricing Archetypes

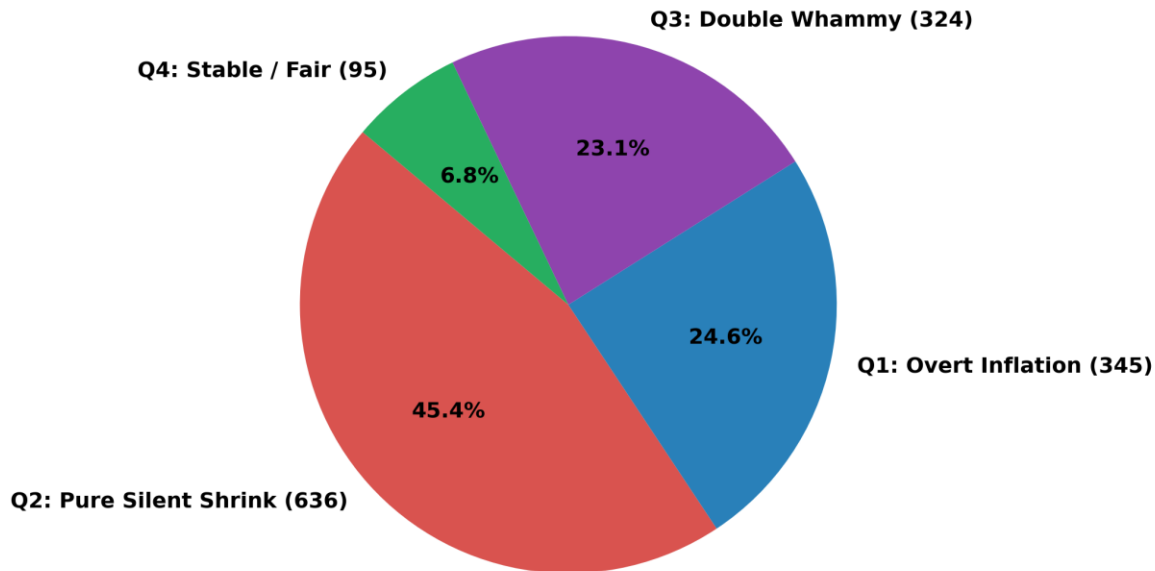


Figure 2. Price–weight quadrant view generated by the project analytics pipeline.

5. Results and Business Findings

5.1 Overall shrinkflation pattern

Using the project's classification rule, 960 of 1,400 SKUs (68.6%) are flagged as downsized. The mean package-weight change among these flagged SKUs is -21.8%. Across all SKUs, the average change in standardised unit price is 31.9%.

Among products at the ₹5/₹10/₹20 price points, 95.1% are classified as shrinkflated, compared with 45.2% of non-magic-price products. The gap is large in this simulated data and supports the project hypothesis that price-point rigidity can be an important feature to investigate.

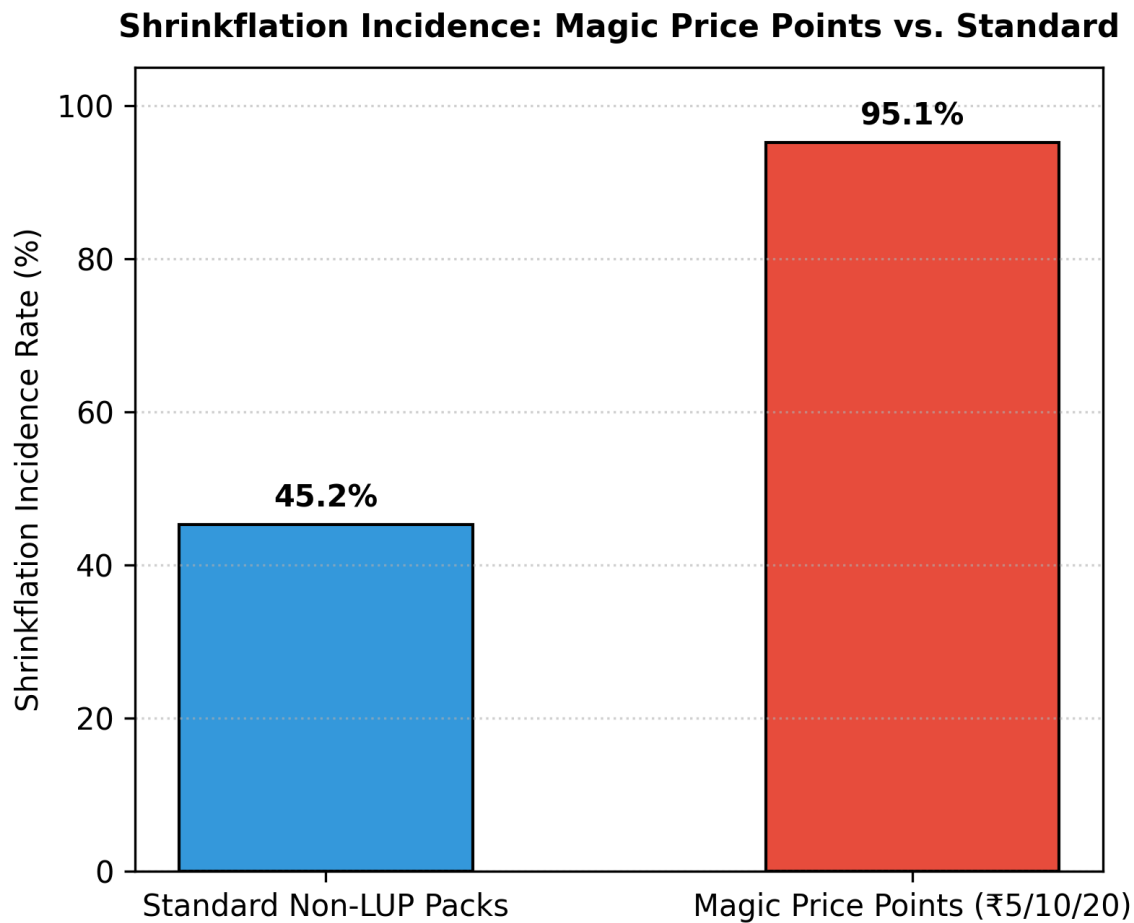


Figure 3. Simulated shrinkflation incidence by low-unit-pack price-point status.

5.2 Four-quadrant results

Quadrant	SKUs	Share
Q2 – Silent Shrink	636	45.4%
Q1 – Overt Inflation	345	24.6%
Q3 – Double Whammy	324	23.1%
Q4 – Stable/Fair	95	6.8%

5.3 Product-level illustrations

Illustrative SKU	Baseline	Terminal	Weight change	Class
Parle-G (Glucose Biscuits)	150.0g @ ₹50.0	109.3g @ ₹50.0	-27.1%	Q2
Maggi 2-Minute (Instant Noodles)	20.0g @ ₹5.0	14.5g @ ₹5.0	-27.5%	Q2
Haldiram's Bhujia Sev	60.0g @ ₹20.0	46.6g @ ₹20.0	-22.3%	Q2
Vim Dishwash Bar (Toilet Cleaner)	18.0g @ ₹5.0	14.0g @ ₹5.0	-22.2%	Q2
Tata Salt Vacuum Evaporated (Iodized Salt)	2000.0g @ ₹205.8	2000.0g @ ₹210.0	0.0%	Q4

These examples are selected from the supplied analytical dataset and are not presented as verified historical brand records. They illustrate how the same framework distinguishes a quantity reduction, a simultaneous price-and-quantity change, and a price-only change.

6. Machine Learning and Predictive Analysis

6.1 Model design

The target variable is binary: 1 if the SKU's package weight falls by more than 3% between the baseline and terminal waves, otherwise 0. The predictors include low-unit-pack status, baseline price, baseline package weight, marketing-redesign claim status, category and packaging type.

The dataset is divided using a stratified 75/25 train-test split. Logistic Regression uses standardised numeric inputs, while Random Forest and Gradient Boosting use the encoded feature matrix. Ten-fold stratified cross-validation is applied to the training set.

6.2 Model evaluation

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	1.000	1.000	1.000	1.000	1.000
Random Forest	1.000	1.000	1.000	1.000	1.000
Gradient Boosting	1.000	1.000	1.000	1.000	1.000

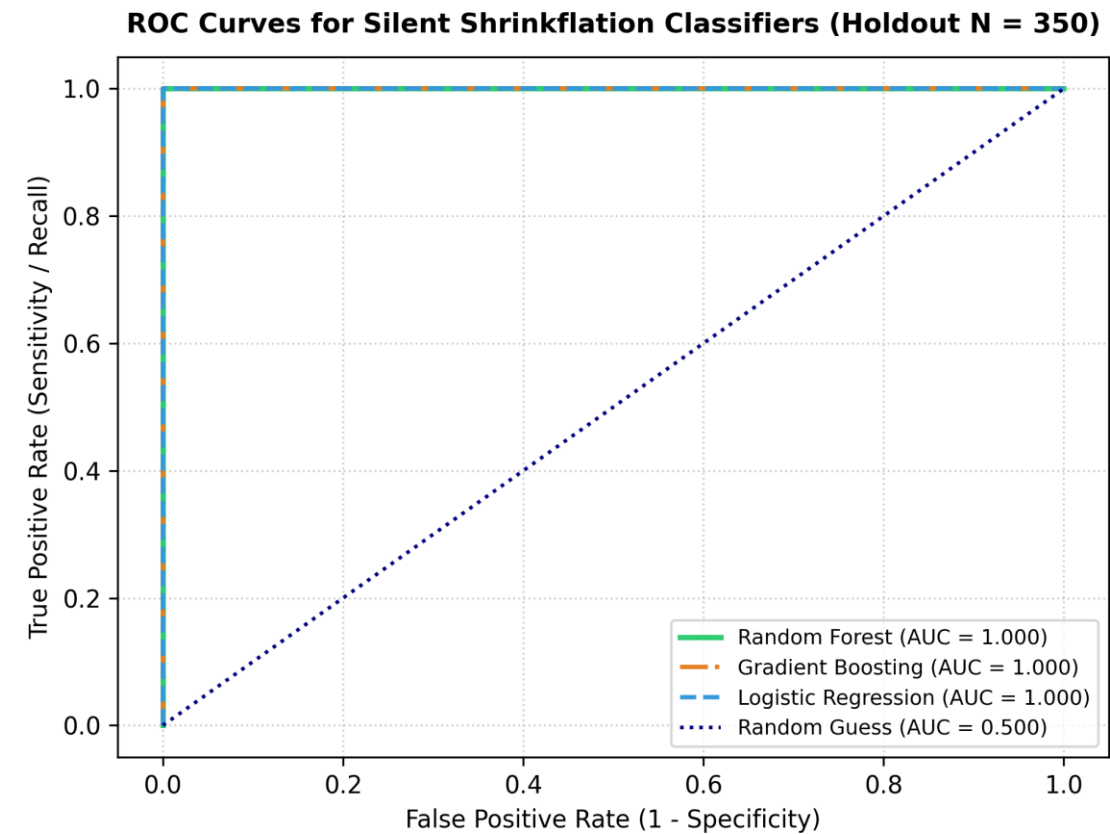


Figure 4. ROC curves for the three shrinkflation classifiers in the current analytical run.

6.3 Feature importance

Feature	Random Forest importance
has_marketing_redesign_claim	0.673
baseline_weight	0.110
is_magic_price_point	0.105
baseline_price_inr	0.062
category_Cooking Staples & Condiments	0.033

In the exact model run contained in the repository, the marketing-redesign indicator has the highest Random Forest importance, followed by baseline package weight, low-unit-pack status and baseline price. This differs from the earlier humanized report, which described magic-price status and baseline price as the main predictors; the present report uses the values from the supplied code and dataset.

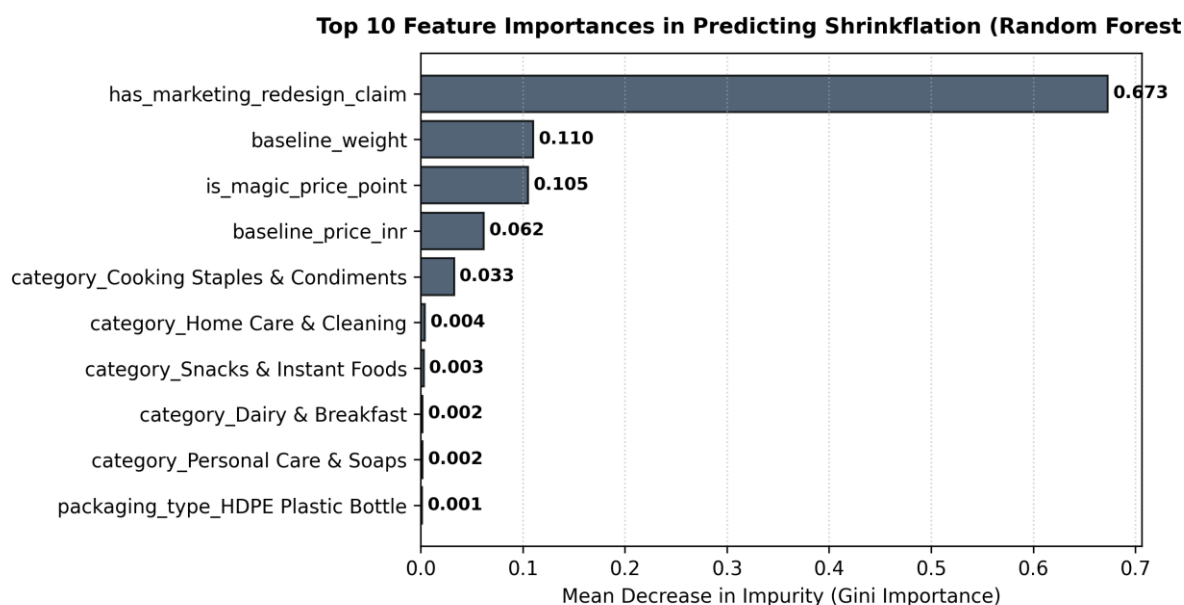


Figure 5. Top Random Forest feature importances from the supplied analytics code.

7. Brand Transparency Index and Business Interpretation

The Brand Transparency Index (BTI) is a project-specific analytical indicator, not an official industry rating. The supplied code combines shrink rate, average Stealth Index and a marketing-claim measure using fixed weights:

$$\text{BTI} = 100 - [45 \times \text{Shrink Rate} + 35 \times \text{Average SI} + 20 \times \text{Deceptive-Masking Rate}]$$

Because the underlying dataset is simulated, company-level BTI values should be read only as outputs of the prototype framework. They should not be interpreted as factual assessments of the named companies.

Company	SKUs	Shrink rate	Avg SI	Unit inflation	BTI
Adani Wilmar	35	11.4%	0.07	+22.6%	90.1
Tata Consumer Products	33	12.1%	0.06	+25.0%	89.9
Procter & Gamble India	21	28.6%	0.17	+22.0%	75.5
GCMMF (Amul)	56	55.4%	0.34	+29.0%	52.0
Mother Dairy	17	52.9%	0.41	+24.3%	51.1
Reckitt Benckiser	50	58.0%	0.37	+29.7%	49.4

Marico	50	64.0%	0.38	+33.0%	44.9
Jyothy Labs	35	62.9%	0.41	+31.3%	44.9
Hindustan Unilever	146	64.4%	0.40	+31.6%	44.1
Dabur India	67	64.2%	0.41	+30.6%	43.9

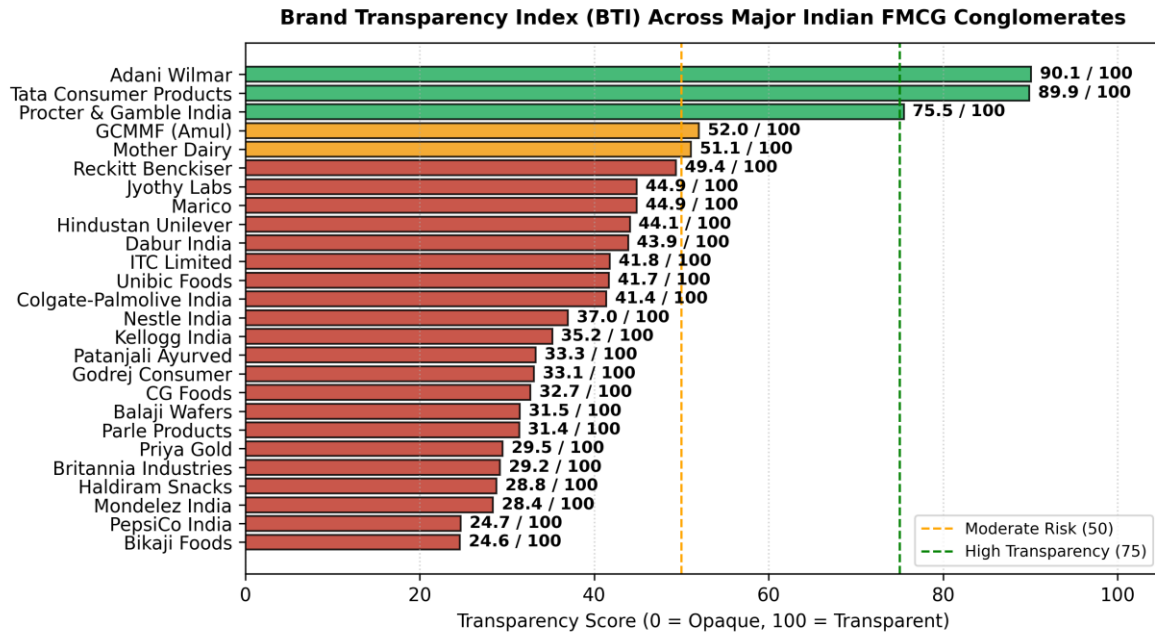


Figure 6. Project-specific BTI output. Values are model outputs on the simulated dataset, not industry ratings.

8. Comparison with Recent Research

Recent shrinkflation research uses several complementary approaches: retail scanner data to measure product-size changes, econometric methods to quantify inflation effects, and experiments to study consumer perceptions. The present project is different mainly in its educational focus on a longitudinal product-level panel, unit-price decomposition, a four-quadrant classification and a machine-learning demonstration.

Study	Dataset	Method	Main finding	Strength	Limitation / difference
Lee (2024)	U.S. store-sales data, 56 categories and 310 sub-categories	Longitudinal package-size and consumer-response analysis	Median package-size reduction of 11%; unit price increased about 9%	Large product coverage and direct consumer-response analysis	U.S. setting; working paper; not focused on Indian quick-commerce
Rojas, Jaenicke & Page (2024)	U.S. packaged-food scanner data, 2012–2021	Hedonic inflation measurement and package-size analysis	Package-size changes can create downward bias in measured food inflation	Connects package size to inflation measurement	U.S. data; conference/working-paper version
Evangelidis (2023/2024)	Five preregistered consumer experiments	Experimental fairness and consumer-perception analysis	Downsizing was often perceived as less fair than equivalent price increases	Directly tests consumer perception and transparency effects	Experimental rather than longitudinal market data
Janssen & Kasinger (2025/2026)	NielsenIQ U.S. retail scanner data, 2010–2020	Product-size changes plus demand estimation	Downsizing was more common than upsizing; consumers responded more to price than size changes	Large-scale market evidence and demand analysis	U.S. scanner data; not an India-specific study
This project	1,400 simulated SKUs × 12 waves	Unit-price decomposition + quadrants + BTI + ML	Prototype identifies simulated downsizing patterns and tests a reproducible analytics pipeline	Combines descriptive, decomposition, scoring and predictive components	Current dataset is synthetic and must be replaced with verified web-scraped observations for a course-compliant primary-data

9. Business Insights and Recommendations

9.1 Insights

- Printed price and effective unit price can tell different stories; pack quantity must be considered alongside price.
- Low-unit-pack products show a much higher simulated shrinkflation incidence than other products in the prototype dataset.
- A four-quadrant framework makes it easy to separate price-only inflation from quantity-only and combined changes.
- Machine learning can be used as a screening tool once a sufficiently large, independently collected dataset is available.
- Company-level transparency indicators can summarise product patterns, but they should be based on verified observations and clearly stated definitions.

9.2 Recommendations for FMCG businesses

7. Monitor price and pack size together rather than treating sticker-price changes as the only pricing signal.
8. Track standardised unit price internally for every major SKU and price point.
9. When a pack is resized, maintain a clear record of the old and new quantity and communicate the change consistently across channels.
10. Use longitudinal dashboards to identify repeated quantity reductions at the same price point.
11. Validate any marketing or packaging claim against the actual quantity delivered to the customer.

9.3 Recommendations for consumers and platforms

12. Display unit price prominently beside pack price on digital grocery listings.
13. Allow shoppers to compare the current pack size with the immediately previous recorded pack size where historical information is available.
14. Flag unusually large quantity reductions for internal review or consumer-information prompts.
15. Use standardised ₹/100g or ₹/100ml comparisons when evaluating products across different pack sizes.

10. Limitations

- The current analytical dataset is simulated rather than verified historical web-scraped data.
- The generated observations are deterministic and therefore do not capture real website volatility, promotions, stock-outs or listing changes.
- The 3% shrinkflation threshold is a project rule, not a legal definition.
- Baseline-to-terminal comparison can miss temporary changes or multiple intermediate size changes.
- The machine-learning target is constructed from the same package-change logic that the models are asked to predict, which can produce optimistic performance.
- BTI is a project-specific indicator and should not be treated as an official brand rating.
- The prototype covers selected packaged-grocery categories and cannot represent the entire Indian FMCG market.

11. Conclusion

This case study demonstrates a complete business-analytics workflow for studying shrinkflation: structured longitudinal product data, unit-price normalisation, price–quantity decomposition, quadrant classification, company-level scoring and machine-learning classification. In the current prototype dataset, 960 of 1,400 SKUs are flagged by the project rule, and the simulated low-unit-pack group shows substantially higher shrinkflation incidence than the standard-pack group.

The main analytical lesson is simple: the sticker price alone is not enough to understand the value received by a customer. A standardised unit price provides a clearer basis for comparison, while longitudinal data make it possible to detect quantity changes that would be invisible in a single snapshot.

For a fully compliant course submission, the next step is not to change the analytical logic but to replace the simulated CSV with genuinely collected public-web snapshots, preserve the raw evidence, rerun the notebook and update the report numbers. That change would turn the current prototype into an evidence-based web-scraping case study.

References

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- Ministry of Consumer Affairs, Food and Public Distribution, Government of India. Legal Metrology overview and Legal Metrology (Packaged Commodities) Rules, 2011. <https://consumeraffairs.gov.in/pages/legal-metrology-overview>
- Government of India, Department of Consumer Affairs. Legal Metrology (Packaged Commodities) Rules, 2011 and subsequent amendments.
- Project repository files: raw_scraped_grocery_data.csv, cleaned_shrinkflation_analytics_data.csv, model_metrics.json, analytics_and_ml.py, scrape_and_generate_data.py.

Appendix A – Reproducibility and Project Mapping

Repository component	Role in the case study
data/raw_scraped_grocery_data.csv	Current 16,800-row simulated longitudinal dataset used by the analytics script.
data/cleaned_shrinkflation_analytics_data.csv	1,400-row SKU-level analytical dataset created from baseline/terminal observations.
data/model_metrics.json	Stored dataset summary, model metrics and BTI outputs.
scripts/analytics_and_ml.py	Performs decomposition, quadrant classification, visualisation, BTI and ML.
scripts/scrape_and_generate_data.py	Current repository script generates simulated observations; it is not evidence of historical scraping.
analysis.ipynb	Notebook artifact for the analytical workflow.
figures/	Generated charts used for the report.
web/	Project web interface files.

Appendix B – Final submission checklist

- Replace the simulated CSV with genuine public-web observations if the course requires primary web-scraped data.
- Keep raw snapshots and record collection dates/times.
- Rerun analysis.ipynb and regenerate all figures and metrics.
- Update every number in this report from the rerun outputs.
- Verify that the README explains the scraping source, procedure, limitations and reproducibility steps.
- Submit the report together with the required notebook, data and repository files.